Language Features under Weight Quantization: Activation Agreement and Intervention Effects

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Abstract

When a language model's weights are rounded to fewer bits, do the features used to explain it still behave in the same way? We study 27 languages, mainly with Gemma-2-2B and a fixed sparse autoencoder (SAE). We reproduce an earlier study and find that omitting the beginning-of-sequence token substantially changes its Gemma measurements. After correcting this input protocol, removing selected language features harms next-token prediction more than removing matched controls in all 27 languages. Gemma-2-9B shows the same broad pattern. Effects remain near full precision at six bits and grow at four bits under simple rounding; GPTQ reduces that change. Activation changes and output effects differ: under legacy RTN at four bits, only 13.2% of a 1,487-feature sample passes the activation-survival cutoff, yet 99.7% can still be identified from their effects. Activation correlation gives a small positive gain in predicting effect similarity. A reading-comprehension test shows no clear effect of the selected groups, and restoring their activations repairs little quantization loss. The study establishes useful effects of specific feature edits and shows why activation agreement, task performance and preservation of an edit's effect need separate measurements.

1 Introduction

Quantization rounds model weights to fewer allowed values. It can make language models cheaper to store and run, but may also change their behavior. The effects can differ across languages and tasks (Marchisio et al., 2024). We investigate what happens to the internal features used to explain these models.

A sparse autoencoder (SAE) is a separate network that describes a model's hidden state using a few active features from a large learned dictionary. Each feature has an activation strength and a direction in the hidden state. Some activate more often for one language than for others. Previous work has identified and edited such language features (Deng et al., 2025); another study measures how their activations change under quantization (Duan, 2026).

We connect these measurements and find a sharp dissociation. At four bits under simple rounding, only 13% of language features pass the activation-survival cutoff, yet 99.7% can still be identified from what they do to the model's output. Activation agreement, task performance and preservation of effect need separate measurements: they answer different questions and give different answers. A changed activation does not tell us whether removing that feature still has the same effect. A feature can matter for predicting the next token without producing a measurable change on a reading-comprehension test.

Four questions organize the completed study:

RQ1: Which language features retain their activation patterns after quantization, and how does the quantizer affect this?

RQ2: How does editing language features affect text prediction and comprehension before and after quantization?

RQ3: How well do activation measurements predict whether a feature's output effect is preserved?

RQ4: How much does the same feature edit change when the model is quantized, and what explains that change?

These final questions organize the accumulated evidence. They do not replace the original hypotheses or the choices fixed before each analysis.

The mechanistic element is the use of SAE features as causal units. Rather than measuring only

whether quantization changes model outputs, we remove, scale and patch individual features and measure whether the same intervention still produces the same downstream effect. This tests where in the model quantization matters and whether the internal explanation survives compression.

2 Related Work

Compression and language. Marchisio et al. (2024) show that multilingual quantization effects vary by task and evaluation. Resource tiers describe differences in language resources (Joshi et al., 2020), but do not isolate how much training exposure a model received. GPTQ uses calibration data to reduce rounding error (Frantar et al., 2022). We compare it with simple rounding under two implementations.

SAEs and direct evaluation. Gemma Scope supplies public SAEs for Gemma models (Lieberum et al., 2024). JumpReLU and TopK provide ways to learn sparse representations (Rajamanoharan et al., 2024; Gao et al., 2024). These tools remain imperfect: features can mix concepts, split them across units, or leave parts of the hidden state unexplained (Leask et al., 2025). Task-based evaluation can still test whether specific edits are useful (Makelov et al., 2025). SAEBench also shows that better proxy scores need not yield better practical performance (Karvonen et al., 2025). We therefore keep the dictionary fixed and measure both activations and controlled edit effects.

Feature preservation and patching. Language-feature editing predates our study (Deng et al., 2025). Related work studies activation survival under rounding (Duan, 2026), geometric survival under pruning (Borobia et al., 2026), and the geometry of feature effects (Hoang et al., 2026). Activation patching can test where quantization error affects outputs (Chang et al., 2025), including interactions between patches (Vaidyanathan et al., 2026). Our contribution is to compare these kinds of preservation across languages and quantizers with explicit control edits. Belebele provides a separate comprehension task (Bandarkar et al., 2024).

3 Experimental Setup

Models and data. The main model is Gemma-2-2B (Gemma Team et al., 2024). Its fixed Gemma Scope JumpReLU SAE has 16,384 features and reads the hidden state after layer 12, near the midpoint of 26 transformer blocks. Gemma Scope provides SAEs at every even layer; we chose the middle layer because prior work found language-direction features there (Deng et al., 2025). Results may differ at other layers. Model inference uses bfloat16; SAE calculations and edits use float32. The Gemma-2-9B extension uses a layer-20, 16k SAE at a similar relative depth in a 42-layer model. A smaller Llama-3.2-1B study (Grattafiori et al., 2024) uses a layer-12 TopK SAE with 14,336 features and $k = 48$. Model, SAE, training data and tokenizer change together in that comparison.

The main panel contains aligned FLORES-200 sentences (NLLB Team et al., 2022) in 27 languages, nine per descriptive resource tier. Feature selection uses the development split. Activation and language-association tests use 27,036 content tokens per language from devtest; prediction-loss tests use 300 aligned devtest sentences. The comprehension test uses 385 Belebele questions per language in 18 languages that passed calibration, grouped into 219 passage clusters. A second transfer panel uses fresh Wikipedia articles in six languages.

Quantization. Round-to-nearest (RTN) maps a weight to its nearest allowed value. Legacy RTN is the original implementation, which shares attention scales across corresponding coordinates of different heads. Matched RTN uses one scale per output channel and the same rounding grid as GPTQ. GPTQ calibrates on 8,192 Wikipedia tokens per language (221,184 total), separate from evaluated sentences and articles. Matching the grid helps distinguish changes in the rounding setup from GPTQ's additional benefit. Calibration and error compensation still change together.

Selected transformer weights are rounded; embeddings and normalization parameters stay unchanged. Values remain stored and evaluated in floating point, so we measure weight-rounding effects rather than deployment speed or memory savings. The original tests cover FP and eight to four bits (INT8–INT4). FP is the unquantized reference. Later intervention comparisons focus on INT6 and INT4. INT16 and INT12 make small real weight changes and serve as reference conditions; INT16 is not identical FP arithmetic.

Analysis status. Most FLORES studies are exploratory: they reuse examined sentences or follow earlier results. Fixing a later stage's choices

before its output does not supply independent confirmation. The original held-out Belebele family is confirmatory. We claim no verified external preregistration. Code hashes, input records, restore checks and sham edits support implementation checks. New quantizer arms reproduce FP reference values exactly. The 9B review regenerates its analysis byte-identically from pinned raw files; it does not repeat GPU inference. This reporting revision adds no experiment or statistical analysis.

4 Problems Found and Corrected

The missing BOS token in the earlier Gemma evaluation. Our first reproduction disagreed with Duan's reported survival rates. Running their released code reproduced their table closely: INT7 survival was 81.19% versus 81.26%, and INT6 was 51.15% versus 51.30%. We traced the large disagreement to a methodological problem in their Gemma input protocol: it omitted the beginning-of-sequence (BOS) token.

Adding BOS to the same 256-content-token WikiText windows, while excluding its position from feature statistics, reduced full-precision perplexity from 458.4 to 16.1. Lower perplexity means better text prediction. INT6 survival rose to 97.52%, and INT7 to 99.97%. Each regime defines its own active-feature set, so these percentages do not follow exactly the same feature population. We corrected our subsequent protocol by prepending one BOS token and excluding BOS and padding from statistics and edits. This demonstrates a large input-protocol effect in the Gemma evaluation; it does not establish that every conclusion in Duan's paper is wrong.

Our own corrections. We retired the first pipeline after finding separate flaws, including counting BOS in feature statistics and changing tokens during decode/re-encode. Its results are excluded. We also corrected an assumption that INT16 was identical to FP: re-rounding makes a small real weight change. A small gain experiment had to be rerun on the intended GPU because the hardware shift was comparable to the measured effect. The 9B review found two defective prediction criteria, described with its results below. The full report and decision log retain these problems and their corrections.

5 Measurements and Controls

The core test is straightforward: for each language, we identify a small group of features that fire selectively for it, remove them, and measure how much worse the model predicts the next token compared with removing matched controls. Doing this at full precision and at each bit width tells us whether quantization changes how much the features matter.

Feature activation and preferred language. A feature fires when its activation is positive. We select language features using the ratio of their highest per-language firing rate to their mean across languages. Selection also requires enough active tokens, a shuffled-language cutoff (50 permutations), and stable peak language in 80 of 100 resamples. This yields 8,086 features. On FP devtest, 6,663 have a unique supported peak language. We test retention of that language only for this established set.

Activation correlation instead compares the same feature's values on the same tokens before and after quantization:

$$R_1 = \mathrm{corr}_t(a_f^{\mathrm{FP}}(t), a_f^b(t)). \tag{1}$$

The historical survival rule is $R_1 > 0.9$. Retaining a preferred language and retaining high correlation answer different questions.

Edits and prediction loss. Each language group contains its 64 selective features with the highest mean development-set activation, called K64. Control groups come from 2,860 supported nonselective features and match firing rate and mean activation on a log scale. Evaluation uses three control draws; the saved manifest stores up to five. Matching is close but imperfect: 75 of 81 language–size ratios fall within $[0.8, 1.25]$, with Thai, Yoruba and Lao outside at K64.

For hidden state x, activation a_f and decoder direction d_f, we edit a group G as

$$x' = x + \sum_{f \in G} (a'_f - a_f) d_f. \tag{2}$$

This retains the SAE reconstruction error. Removal sets $a'_f = 0$, scaling sets $a'_f = \alpha a_f$, and a donor patch uses the FP activation at the same position. For example, Turkish has 64 selective features; setting all 64 to zero and measuring the increase in

next-token loss for Turkish sentences, minus the increase from zeroing a matched non-selective group, gives the excess harm E.

Prediction loss, or negative log-likelihood (NLL), measures how much probability the model assigns to the actual next token. Lower is better; units are nats/token. We subtract control-group harm from language-group harm and call the result excess harm, E_ℓ^b. Positive E means the language edit hurts more. The effect ratio $E_\ell^b / E_\ell^{\mathrm{FP}}$ compares it with FP: one means equal harm, 1.5 means 50% greater harm. The median of language ratios, their mean, and the ratio of mean effects are reported separately because small denominators can make them differ. Patch recovery measures how much of the loss gap a specified restoration removes.

Output-effect similarity and uncertainty. A feature's effect fingerprint is the vector of changes in output-token scores caused by removing it. Cosine similarity with its FP fingerprint, W_1, measures how closely those directions agree; one means identical direction. The F5 study samples 1,487 features; 1,412 pass the INT16 reliability criterion. A stricter similarity cutoff, 0.9489, comes from the fifth percentile of INT16 cosines. Close agreement with this reference and recognizable feature identity are different tests.

Bracketed ranges are 95% bootstrap intervals unless stated otherwise. Resampling units follow the experiment: paired sentences for group loss, passages for Belebele, languages for F5 prediction and reviewed 9B comparisons, and features plus source sentences for transfer. Intervals hold other choices, such as groups and controls, fixed. They do not describe uncertainty over all model or SAE choices.

6 RQ1: Which Activations Survive?

Feature activity changes the comparison. The selective-vulnerability hypothesis from our initial study was not supported after correction for multiple comparisons (adjusted $p = 0.514$ at INT7, 0.128 at INT6). The stronger evidence throughout the study comes from the consistent direction of effects across 27 languages and from bootstrap intervals, not from individual significance tests. Survival was lower than controls by 0.28 points at INT7 and 1.44 points at INT6. Small effects remain possible. A stricter selectivity cutoff ($S > 5$) gives a larger INT6 gap of -7.16 points; matching activation magnitude in ten bins reverses it to $+4.58$.

INT4 measurement	Legacy	Matched	GPTQ
Active-feature survival (%)	12.7	18.6	30.5
Mean loss increase (nats/token)	0.590	0.385	0.130
Median group-effect ratio	1.54	1.36	1.22
Median effect cosine (W_1)	0.839	0.875	0.923
Transfer cosine	0.768	0.822	0.877
Belebele accuracy (%)	23.6	29.1	31.8

Table 1: Completed Gemma-2-2B quantizer comparison. Rows answer different questions. Survival averages active-feature shares per language; W_1 uses F5 features; transfer uses eligible feature–context pairs; Belebele uses 18 languages.

Low-, mid- and high-resource languages have INT6 survival rates of 69.4%, 90.6% and 95.1%. Accounting for activation scale reduces the fitted tier slope from $+12.3$ to -0.81 points. Later measures still show some tier patterns at INT16, and GPTQ's adjusted result depends on the adjustment choice. These measurements alone do not identify a training-resource mechanism behind language damage. Individual languages deviate from tier averages: some low-resource languages have more selected features and larger FP excess harm than some mid-resource languages, consistent with feature count reflecting the model's internal specialization rather than training-data volume alone.

GPTQ improves preservation. At INT6, average active-feature survival is 85.0%, 89.7% and 97.5% for legacy RTN, matched RTN and GPTQ. At INT4, GPTQ exceeds legacy RTN by 17.83 points $[16.29, 19.38]$ and matched RTN by 11.86 $[11.18, 12.50]$. It also reduces the mean loss increase by 0.461 nats/token $[0.348, 0.595]$ relative to legacy RTN. Both outcomes improve in all 27 languages. Intervals resample paired languages. Matched RTN closes part of the gap by itself. Its small selective-feature contrasts cross the significance threshold, while GPTQ's do not. These exploratory comparisons support a strong quantizer effect and no consistent selective penalty across all three methods.

Correlation misses some activation changes. Multiplying every activation by the same factor can leave correlation unchanged. We therefore decompose quantized activation as a scaled FP signal plus a residual: $b = ga + c + e$. Here g is gain, c

is an offset, and e is the part scaling does not explain. The residual is uncorrelated with a. Where the quantities are defined,

$$\lambda = \log(R_1^{-2} - 1) = \log \sigma_e^2 - 2 \log |g| - 2 \log \sigma_a. \tag{3}$$

The correlation cutoff 0.9 corresponds to $\lambda < -1.450$, with positive gain. We also measure preactivation change divided by the feature's learned threshold, denoted ν, on tokens active in FP. This puts the change in threshold units.

The audit covers 214,506 active language–feature pairs. It reproduces prior correlations within 3×10^{-8} and checks the identity to about 10^{-14}. Across six INT4/6 conditions, rank correlations with FP activation spread are -0.507 to -0.534 for λ, versus -0.004 to 0.114 for ν. Threshold scaling reduces this dependence, though dependence on peak activation remains. These measures explain different parts of activation change; the prediction test below asks whether they add useful information.

7 RQ2: What Do Language-Feature Edits Change?

Removing features harms next-token prediction. FP excess harm is positive in all 27 languages of Gemma-2-2B: mean 0.704 nats/token $[0.684, 0.723]$, median 0.262. The effect stays near FP at INT6, then grows at INT4. Legacy INT4 has a median ratio of 1.54 $[1.43, 1.62]$, a mean ratio of 2.14 $[2.04, 2.50]$, and a ratio of mean effects of 1.33. GPTQ lowers the median to 1.22 $[1.14, 1.24]$. Its mean ratio is 1.31 $[1.27, 1.50]$, leaving its classification against the predefined $[0.75, 1.33]$ band uncertain. Greater removal harm indicates greater sensitivity to the edit.

Gradual suppression supports the same result. In all 27 languages, FP harm increases as the activation multiplier decreases. Median harm is 0.0223, 0.0902, 0.1687 and 0.2624 at multipliers 0.354, 0.177, 0.088 and zero. The strict rule requiring monotonic change across every precision fails because of a small intermediate-width dip; the suppression-dose and consistency rules hold.

A second Gemma size and a different setup. Gemma-2-9B shows positive FP excess harm in all 27 languages, with median 0.270 $[0.197, 0.608]$. The median effect ratio is 1.019 $[0.978, 1.045]$ at INT6 and 1.586 $[1.238, 2.264]$ at INT4. At INT4, the mean ratio is 2.479 and the ratio of mean effects is 1.474. These intervals resample languages. Paired 2B–9B difference intervals include zero, so the data support the broad pattern across sizes without resolving a size effect.

The 9B prediction checklist had two defects: its suppression inequality was reversed, and its selectivity criterion was guaranteed by how features were selected. We do not count those as independent confirmations. The measured suppression-harm ordering holds in all 27 languages. At INT4, GPTQ lowers the median effect ratio to 1.165 $[0.997, 1.370]$, below legacy RTN in 26 of 27 languages, and matched RTN gives 1.490. Activation survival rises from 22.3% (legacy) to 32.9% (matched) and 46.6% (GPTQ). Mean loss increase falls from 0.410 to 0.287 and 0.088 nats/token. All frozen predictions for both tiers were met.

Llama has a smaller positive FP median effect, 0.048 $[0.018, 0.173]$, with positive effects in 21 of 27 languages. This misses the consistency rule requiring 24. Its INT4 median ratio is 0.86 $[0.74, 1.03]$, with less consistent suppression ordering. The effect differs across these model/SAE setups.

The comprehension effect is not established. On held-out Belebele, FP excess correct-option loss is -0.0001 nats/item $[-0.0035, +0.0031]$. Accuracy is about 42.6% unedited, 42.6% after language-group removal and 42.5% after control removal. The required positive FP effect is absent, so the five planned conditional retention statements cannot be interpreted. Larger nonselective edits (1,024–2,860 features) lower accuracy by 1.2–2.2 points; other-language selective edits reach 6.0 points. Sham edits change nothing. GPTQ's INT4 baseline accuracy is 31.8%, above 25% chance, and the K64 effect remains small. The test detects larger edits but leaves small K64 effects possible.

Repair gains are limited. Restoring FP activations in K64 recovers 2.6% $[2.4, 2.8]$ of the legacy INT4 loss gap, versus 39.3% $[38.5, 40.0]$ for the full residual. GPTQ gives 2.0% $[1.6, 2.3]$ versus 41.4% $[40.2, 42.4]$. These are measured patch benefits, not additive shares of the cause of model damage.

The main fine-gain study fails its repair rules. A legacy-RTN follow-up on 562 fresh sentences finds a small benefit at multiplier 1.1: -0.00135 nats/token, with 97.5% interval $[-0.00197, -0.00032]$, and improvement in 20 of 27 languages. It meets the rule after multiple-

comparison correction at 1.1, but not 1.2. The primary pair of multipliers uses Holm correction. GPTQ does not repeat the fine-gain trend on the main 300-sentence panel; the fresh-sentence test was legacy-only.

Selecting groups anew after quantization changes their membership (overlap with FP falls from Jaccard 0.855 at INT6 to 0.488 at INT4), but does not make removal more harmful (median difference -0.0098). The most effective activation multiplier is repeatable within the study, yet does not track inverse activation shrinkage (Spearman $\rho_s = -0.17$, $p = 0.804$). Simple shrinkage compensation is therefore not supported by this test.

8 RQ3: Do Activation Measures Predict Output Effects?

Of the 8,086 selected features, 6,663 establish a peak language on FP devtest. At legacy INT4, 70.8% retain it, while only 13.75% in this group pass activation correlation 0.9. Near-tied FP language preferences are more likely to switch: peak-margin discrimination scores are 0.90, 0.81 and 0.75 at INT6, INT5 and INT4. Removing matched lost-language and retained-language subgroups gives little harm difference: medians $+0.0006, +0.0007, +0.0002$ nats/token at INT4/5/6.

The individual-effect experiment shows that activation change can coexist with recognizable function. In the same 1,487-feature sample, only 13.2% pass the activation cutoff at legacy INT4, but median output-effect cosine is 0.839 $[0.829, 0.848]$. Most effects remain distinctive: 99.9% exceed different-feature controls and 99.7% identify their own FP feature. Only 0.3% meet the stricter INT16-based similarity cutoff. GPTQ improves median cosine to 0.923 and raises the share passing that cutoff to 11.7%. Keeping the same language label does not reliably identify which effects meet the preservation criterion; discrimination scores are near chance (legacy 0.506, GPTQ about 0.49).

Continuous activation correlation is more informative. On 1,412 reliable features, we predict effect similarity while holding out one language at a time. Adding correlation to activity, context-count and FP-effect-size predictors reduces mean absolute error by 3.1–4.4% across all six quantizer/width conditions. Every language-bootstrap interval is above zero. At INT4, the gain is 3.7% $[2.3, 5.2]$ for legacy RTN and 3.5% $[2.1, 5.0]$ for GPTQ. These are small positive gains below the project's 10% practical target. Partial rank correlations are 0.239–0.399.

Adding gain and threshold-scaled change ν after correlation gives further improvements of -0.4 to $+1.8$%, with every interval including zero. The measures help describe activation changes without showing an additional prediction benefit. Llama's 610 reliable features show a similar INT4 gain, 4.2% $[1.6, 6.7]$, with partial correlation 0.252 $[0.184, 0.322]$; INT6 is uncertain.

9 RQ4: Does the Same Edit Still Have the Same Effect?

Separating earlier context from later weights. We fix the edit direction and strength at their FP values. We then cross FP and quantized earlier layers with FP and quantized later layers. Let $D[p, s]$ be the resulting output-logit change, subtracting each condition's own unedited output. With Q denoting quantized,

$$C = D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{FP}], \tag{4}$$

$$W = D[\mathrm{FP}, \mathrm{Q}] - D[\mathrm{FP}, \mathrm{FP}], \tag{5}$$

$$I = D[\mathrm{Q}, \mathrm{Q}] - D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{Q}] + D[\mathrm{FP}, \mathrm{FP}]. \tag{6}$$

C measures the context contribution, W the later-weight contribution and I their interaction. The vectors add exactly; their lengths and medians do not. For groups, we compute the same crossed contrasts directly in prediction loss. A fifth condition uses the quantized model's own activation strength.

The panel has 404 features and 6,464 feature–context pairs. We retain 2,203 pairs from 235 features whose FP effects exceed five INT16 reference changes. A matched different-feature direction and three random directions with equal norm test sensitivity shared with other edits. Prediction holds out feature identities, while source sentences may occur in both splits.

At legacy INT4, median group excess harm is 0.848 nats/token for the fixed-strength edit, versus 0.263 in FP and 0.433 using native quantized strength. Median absolute context and weight terms are 0.310 and 0.104 nats/token. Native-minus-fixed harm is negative in every language, with median -0.351 $[-0.358, -0.293]$. The native-minus-weights-only remainder is smaller in the median language ($+0.0138$ $[0.0093, 0.0233]$), with variation across languages. For individual output vectors, the weight term is larger. Attribution

therefore depends on both the chosen edit and the outcome.

GPTQ transfers the edit more closely. At legacy INT6, output-effect cosine is 0.984 and relative error is 0.165, about 1.88 INT16 reference floors. At INT4, cosine is 0.768 $[0.760, 0.776]$ and relative error 0.664. Matched RTN improves cosine to 0.822 and GPTQ to 0.877; relative error falls to 0.572 and 0.476. Lower relative error means the same edit gives a more similar output change.

Control directions also drift substantially. A later comparison matched FP effect size on 144 FLORES pairs and 107 Wikipedia pairs. Real-direction INT4 error norms are 1.14 $[1.07, 1.25]$ and 1.18 $[1.05, 1.24]$ times the swapped-control norms. This shows a modest extra component on the selected subsets, alongside a large shared component.

Using control-direction sensitivity improves held-out legacy INT4 error prediction by 21.5% $[12.7, 29.4]$. At INT6 the gain is 10.9% $[-1.2, 21.4]$, so the original rule requiring utility at both widths is not met. A similar INT16 gain suggests sensitivity to perturbations generally. Activation survival and the scalar-loss gradient add no clear prediction gain for this vector-error target. The gradient does predict its own loss target moderately ($r = 0.63$ at FP, 0.58 for the INT4 shift); we did not compute a vector-gradient approximation.

Averaging effects across contexts makes preservation look stronger. For legacy INT4, cosine after averaging is higher by 0.126 than the average of context-level cosines. Reports should say which comparison they use.

Fresh text gives a similar INT4 pattern. On the six-language Wikipedia panel, INT4 cosine is 0.754, 0.808 and 0.876 for legacy RTN, matched RTN and GPTQ. Control-direction sensitivity improves INT4 prediction by 19%, 29% and 30%. At INT6 it meets the utility rule for both RTN methods, but not GPTQ (3% $[-14, +17]$), whose error is near the reference scale. GPTQ calibration uses separate articles from the same domain, which may contribute to its advantage.

10 Discussion and Conclusion

The central finding is a dissociation: activation survival and functional preservation come apart. At four bits, 87% of language features look "dead" by the standard activation cutoff, yet almost all still produce recognizable output effects. A direction-swap control shows that the similarity loss at INT4 is generic to downstream computation, not specific to the features themselves. And restoring the features' original activations recovers only 2.6% of the quantization damage, meaning the amplified edit sensitivity does not point to the features as the locus of harm. These three results together say that the features survive functionally but carry almost none of the model's loss from compression.

The selected groups contribute to next-token prediction in all 27 languages on both Gemma sizes, with dose–response confirmed in every language. GPTQ roughly halves the INT4 amplification and the completed 9B three-arm comparison shows the same quantizer ordering as 2B, with all frozen predictions met. The Llama check, where only 21 of 27 languages show a positive effect, bounds the finding within the Gemma family.

Other tests show where the interpretation breaks down. The selected groups show no clear effect on the reading-comprehension task, and the initial selective-vulnerability hypothesis was not supported after multiple-comparison correction. These negative results are as informative as the positive ones: they mark where the proposed explanation holds and where it does not.

Limitations, Challenges and Future Work

Most evidence comes from Gemma-2-2B at one layer and one fixed SAE. The 9B three-arm extension and narrower Llama check broaden the evidence, but results may differ with other models, layers or dictionaries. Features can mix or split concepts, and large edits can create unusual hidden states. Activity matching and direction controls reduce some differences without matching every feature property. The results apply to the specified edits and outcomes.

Most FLORES analyses reuse examined sentences. Holding out languages or features does not necessarily hold out documents. Intervals condition on the feature groups and other fixed choices. Language tiers also mix script, tokenization, training exposure and competence. Truncating inputs to a fixed token budget removes the length confound but may alter internal representations compared with complete sentences; matching sentence length would trade truncation for a sentence-complexity confound, and varying both independently would require a corpus that FLORES-200 alignment does not provide. Belebele tests one comprehension

task, while the external Wikipedia panel has six languages and shares a domain with GPTQ calibration.

The quantization experiment rounds weights during floating-point inference. It does not measure integer-kernel deployment or activation quantization. Calibration and error compensation are not varied independently. Small results can depend on hardware and arithmetic precision, as the correction record shows. Additional independent text, dictionaries and precision comparisons would test broader generalization.

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A Language Panel and Reproduction

Tier	FLORES-200 language codes
High	eng_Latn, deu_Latn, spa_Latn, fra_Latn, por_Latn, rus_Cyrl, zho_Hans, jpn_Jpan, kor_Hang
Mid	hin_Deva, ben_Beng, tam_Taml, tur_Latn, vie_Latn, ind_Latn, tha_Thai, pol_Latn, ukr_Cyrl
Low	kan_Knda, tel_Telu, yor_Latn, sun_Latn, amh_Ethi, ibo_Latn, hau_Latn, khm_Khmr, lao_Laoo

Table 2: Fixed descriptive tiers, nine languages each.

The repository is https://github.com/TH4KUR/always-late; committed evidence maps identify result files, freezes and hashes. The artifact dataset is https://huggingface.co/datasets/karma-skz/always-late-artifacts; access is private and not all item-level files are public. Run summaries are at https://wandb.ai/karma-skz-/anlp-project. The current evidence map covers the whole completed study. Earlier report wording is archived verbatim.

The frozen environment records Python 3.10, PyTorch 2.14.1, TransformerLens 3.9.0, SAE Lens 6.54.5 and Transformers 5.19.0. Most 2B jobs used 11 GB RTX 2080 Ti accelerators; the supplied 9B run used a rented 96 GB RTX PRO 6000. Checks include sham identities, restored FP weights, quantized-weight hashes and analysis reproductions. The 9B review checks raw-file provenance and headers and reproduces tables byte-identically; it does not independently repeat the supplied GPU inference. All numbers here derive from completed saved outputs; this revision introduces no model run or new statistical analysis.

B Original Test Rules and Outcomes

Table 3 lists the original test families and outcomes. Its codes link the results to the analysis records. The final RQs organize the completed work without replacing these tests. A small positive estimate remains a result even when it misses a practical target.

Block	Rule or question	Completed outcome and status
Initial contrasts	Three selective-survival/tier contrasts; Holm within family	No rejection under original BOS-masked RTN. Matched/GPTQ reruns are exploratory additional families.
Association	FP establishment and peak retention	6,663/8,086 established; 70.8% retain peak at legacy INT4. Margin prediction is descriptive.
Functional F1–F4	Excess group loss, donor recovery, retained/lost subgroup effects	27/27 positive FP group effects; INT4 K64 recovery 2–3%; small subgroup gaps. Exploratory.
Belebele	Positive FP effect required before five precision statements	S0 not met; S1–S5 undefined. Original held-out family confirmatory; whole-group/prompt/positive-control checks exploratory.
Study G	H-G1/2 suppression dose/specificity; H-G3 boost dose; H-G6 consistency	Met under legacy RTN. Strict precision ordering H-G4 not met; a two-error tolerance sensitivity does not alter the original verdict.
Repair and gain	H-G5 generic/group-specific repair and inverse-shrinkage optimum	Main criteria not met. Fresh-sentence extension C meets its $1.1\times$ rule only; D shrinkage relation not supported.
State-defined G	Groups move; new groups more consequential; dose and repair	Group movement, dose and group-specific repair rules met; greater consequence and generic repair rules not met. Exploratory.
F5 and prediction	Identity, effect drift, association discrimination; continuous R_1	High identity retrieval despite drift. Binary association inconclusive/non-discriminative; continuous gain 3.1–4.4%, positive intervals, below 10% target.
Measurement audit	Gain/noise identity, scale separation, adjusted tiers and extra predictors	Identity verified; scale separation met by convention. Tier estimates mixed; gain/ν increments include zero. Exploratory.
Transfer	Fixed edit; H1 residual drift, H3 prediction, H4 averaging; group A3	Main H1/H4 supported; H3/A3 mixed; combined utility not met. External RTN utility met; GPTQ INT6 unresolved. Exploratory.
Llama	Frozen existence/dose/INT4 consistency rule	Not consistent (P1 21/27, 24 required); FP median effect positive. INT4 F5 association/prediction modestly positive. Exploratory.
Gemma-2-9B	FP effects, precision ladder, three-arm comparison	Positive 27/27, INT6 near FP, INT4 increase. GPTQ lowers median ratio to 1.165; all frozen predictions met. Two defective criteria documented.

Table 3: Completed study inventory. Outcomes follow each test's original rule. The main text and evidence map give estimates, uncertainty and explanations.